

process as described above. This layer acts as an insulating layer for successive layers.

Photolithographic process-2

Application of photoresist layer-2. A UV-dissolvable photoresist is coated over the wafer on the entire surface as shown in Fig. 8.

Application of layer-2 mask. A UV mask resembling the layout of the corresponding layer is applied over the photoresist as shown in Fig. 8.

When a UV ray is passed for about 10 minutes, the mask creates an impression on the photoresist through the opened areas which corresponds to the circuit's layout. The photoresist on the areas where UV rays fall weakens and can be removed by dissolving in a solvent for about 30 minutes. Now the SiO_2 layer is visible though the open-

or tungsten is deposited on the entire outer surface of the wafer for providing metal contacts as shown in Fig. 11.

Metal contacts are removed from places where no external connections are needed. Next, drain, source and gate contacts are created by surface polishing and metal evaporation. The areas where metals are applied and visible, are called 'contact pads.' These contact pads temporarily act as pins where voltages and currents can be applied and verified for testing purposes. The resulting wafer is shown in Fig. 12.

Wafer processing

The wafer of the MOSFET device so fabricated needs to be tested for compliance to the design criteria. Wafers are usually prepared circular in shape as many of the wafer preparation

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ings and hidden elsewhere.

Oxide etching. The visible SiO_2 area represents the layout. These SiO_2 layers are removed by a process called de-oxidation. This leaves the substrate prepared for the second layer of impurity deposition (refer Fig. 9).

Impurity deposition and de-oxidation

Now the pentavalent impurity is diffused over the substrate to form the channel of the MOSFET on mid portion as shown in Fig. 10.

Photolithographic process-3

The aforesaid process is repeated to create gate contacts by metal deposition techniques.

Metal deposition and evaporation

About $10\mu\text{m}$ thick layer of aluminium

steps, viz, photoresist application and impurity diffusion, are highly dependent on centrifugal forces that are possible only in circular motions. These normally consist of thousands of units called dies which eventually become integrated circuits. Wafers are highly fragile and vulnerable to electrostatic discharges and moisture. Therefore these need to be handled with extreme care.

Wafer testing

Instruments like wafer probers are used to test the wafers by applying suitable voltages and currents and measuring back the outputs from the contact pads. Bad wafers are segregated by putting a black dot by the prober machines and eventually discarded. The wafers that are tested and verified to be good, are taken to further processing like packaging of the dies.

IC packaging

Packaging protects dies from change in functionality caused by ambient moisture. It also provides mechanical support, thermal dissipation capability and sturdiness to the device, so the device can be handled by the end users without worrying about its safety.

During packaging, aluminium or tungsten wires are bonded to the contact pads and taken out for external contacts to the metal frame. This process called 'wire bonding' is performed in moisture-free environments. Heat dissipation capability is provided from the bottom of the dies through the package layers to the external environment. If heat dissipation mechanism is not properly designed, the ICs will fail invariably—a condition known as premature death of ICs.

IC package testing

The packaged devices are tested for adherence to design specifications. The failing packages are discarded. The qualifying packages are classified according to the number of tests they pass. The test report is analysed for possible feedback on improving the processes or used for validating the processes.

Volume production and quality assurance testing

The manufacturer can produce millions of these units, each of which undergoes the same testing processes in turn. The devices found to be working satisfactorily are shipped with lot number, batch number and other identification data. A sample taken out of the bulk of these shipments is subjected to quality assurance testing. If the sample device fails, the entire lot is sent back for re-testing. The current technology aims at achieving six sigma level, which means the failing device is expected to be one in a million. ●

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